pfi

Python for Fixed Income

COURSE 8 · IMPLEMENTING AGENTIC AI

Document Tools

WEEK 2 · DAY 2

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Today: documents as tools

Six tools, every text with its source line

`statement_dates`

`statement_text`

`list_sections`

`read_section`

`eightk_section`

`search_indenture`

Today

1. Documents as tools
2. The body, not the contents
3. Search as a tool
4. Quotes and citations, checked
5. NOT IN TEXT, through a tool
6. An 8-K section, described
7. What a long result costs

Every model response in the notebook is a saved response, replayed. A live run can differ.

The rules for documents

An FOMC statement: what it states, never a policy view

Allowed names	Never called
the statement states	hawkish or dovish
a sentence in one and not the other	a policy view
the date of the statement	what the Committee will do

An indenture: what it states, never a judgement of the issuer

Allowed names	Never called
the indenture states	loose, tight, weak, strong
extracted value, with its quote	covenant-lite or protective
passed the source check	a credit view or recommendation

Documents as tools

statement_text("2026-09-16", 300): the text, the cut marker, the source line

```
The Committee decided to raise the target range for the federal funds rate by 1/4 percentage po...

... [cut: 843 characters in all]
```

Source: FOMC post-meeting statement of 2026-09-16, Board of Governors of the Federal Reserve System, https:...

```
A date with no statement: {'error': 'no statement dated 2026-09-17 in the file'}
```

a dict

the same shape from every tool

max_chars

the model chooses how much text

Source:

the last line of every text

A quote the model copies always arrives with the name of its document.

The body, not the table of contents

Every heading textkit's pattern reads as Section 4.12 in Boyd's text

```
character   1,996:  Section 4.12. Limitation on Indebtedness 56

character 198,454:  Section 4.12. Limitation on Indebtedness.
```

find_sections keeps the heading at 198,454, the last one, of 107 sections

56

a page number: the contents line

11,868

characters in the body of 4.12

2.0 to 1.0

the ratio, in the body's first sentence

Source: Boyd Gaming Corporation, indenture for the 4.750% Senior Notes due 2031, Exhibit 4.1 to the 8-K filed 2021-06-08, accession 0001193125-21-185495, Section 4.12.

read_section reads from the body's heading: the section's words, not its page number.

Search as a tool: ranks, never scores

search_indenture("limitation on liens", 3) over 454 chunks of 384 numbers

```
rank 1  Section 4.14   Section 4.14. Limitation on Liens. The Company shall not, and shall no
rank 2  Section 4.14   Liens (or any portion thereof) described in the definition of "Permitt
rank 3  Section 1.01   arrangements for the sale of goods entered into by that Person or any
```

Source: Boyd Gaming Corporation, indenture for the 4.750% Senior Notes due 2031, Exhibit 4.1 to the 8-K filed 2021-06-08, accession 0001193125-21-185495.

no score

last digits move with threads

in the key

a result is part of every later request

> 1e-6

every gap the builder checked

A rank is how close the words sit. It is not a judgement of the text.

Quotes and citations, checked

citations_ok (Course 7's judge) and quotes_found, on the saved answers

run	accepted	citations	numbers	missing	quotes found
fomc	False	False	False	no section read [True]	
section	True	True	True	[]	[True]
search	False	True	False	['2021']	[True]

moved to [Section 4.10]: {'numbers_ok': True, 'citations_ok': False, 'accepted': False, 'missing': []}

"the Company or any Guarantor may Incur Indebtedness if the Company's Consolidated Fixed Charge Cover...

search: missing 2021, the digits of boyd_2021. A pattern on digits cannot tell a file name from a figure.

NOT IN TEXT, through a tool

What the agent did, step by step

```
1  read_section  {"indenture":"boyd_2021","number":"4.12","max_chars":6000} tool_calls 762 31
```

The saved answer

```
The section read (4.12) is about "Limitation on Indebtedness" and does not mention yield to maturity.

Therefore, the yield to maturity is NOT IN TEXT.
```

Model gpt-4.1-mini-2025-04-14, temperature 0, seed 42; 2 requests, 2,991 prompt and 100 completion tokens; saved run recorded 2026-10-11; replayed (no key loaded).

The facts hold; the reply asked for is NOT IN TEXT. An explanation that ends with it is counted apart.

An 8-K section, described

Item 2.03, as its filer wrote it: the saved answer's quote

```
The section states: "On June 15, 2021, Public Service Company of New Hampshire, doing business as...

$350,000,000 aggregate principal amount of its 2.20% First Mortgage Bonds, Series V, due 2031
```

The answer wrote the inner quotes as \"Company\"; the quote check reads them as plain quotes.

```
quotes found [True, True]; numbers no tool returned []
```

Source: PUBLIC SERVICE CO OF NEW HAMPSHIRE, Form 8-K filed 2021-06-21, accession 0001104659-21-082904, Item 2.03.

Model gpt-4.1-mini-2025-04-14, temperature 0, seed 42; 2 requests, 1,999 prompt and 230 completion tokens; saved run recorded 2026-10-11; replayed (no key loaded).

What the filer wrote, under the item it chose. Not a view on the bonds, not a view on the issuer.

What a long tool result costs

Prompt tokens by request, in the saved runs of the lesson

run	tokens by request	added	tool result characters
fomc	762, 1053	291	1109
section	783, 1372	589	2353
search	783, 1107	324	1203
yield	762, 2229	1467	6368
eightk	783, 1216	433	1379

589

tokens added: Section 4.12 at max_chars 2000

1467

tokens added: Section 4.12 at max_chars 6000

Every later request carries the whole transcript: a long result is paid for again at every step.

Excel and Git

Excel side by side (tested in Excel)

`=ISNUMBER(MATCH(B2,C2:G2,0))`	`B 4.12 (text)  C 4.12  (section)`	TRUE
`=ISNUMBER(MATCH(B3,C3:G3,0))`	`B 4.14 (text)  C 4.14  (search)`	TRUE
`=ISNUMBER(MATCH(B5,C5:G5,0))`	`B 4.12 typed as a number`	FALSE
`=C8-C7`	`prompt tokens, section run`	589
`=C11-C10`	`prompt tokens, yield run`	1467

Excel against Python: 4 of 4 rows the same

```
...at least 100, for example [-6000.-]{+2000.+}
first request's key: b96043f99924 saved, dfdab26f5767 edited
```

git diff --word-diff: one word of a tool description, and every request key changes. 37 passed.

The AI check

The docstring is the prompt. The agent code that comes back runs and looks right.

1

Docstring

Read the section from the body: its first sentence, never the contents line.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

The text for 4.12 must hold Incur any Indebtedness. Run it: it fails.

→

4

Fix

Let textkit.find_sections find it: the last heading, the body's.

Today: text.find returns the first heading, the table of contents line, so the reader returns no covenant.

Code that runs is not the same as code that is right.

What you can do now

- Documents as tools, every text with its source line
- Read a section from the body, not the contents
- Search as a tool: ranks and texts, never scores
- citations_ok and quotes_found; read the missing list
- NOT IN TEXT through a tool; an 8-K described

Week 2, Day 3 MCP: the same tools, served by a server